AI CAREERS FOR WOMEN
LINKEDIN FELLOWSHIP

Netflix Performance and Engagement Dashboard

Project Domain

Excel / Power BI

A Project Report

submitted in partial fulfilment of the requirements of the AICW Project

by:

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Year - 2026

ACKNOWLEDGEMENT

We would like to take this opportunity to express our deep sense of gratitude to all individuals who helped us directly or indirectly during this project work.

Firstly, we would like to thank our supervisor, Umamaheswari, for being a great mentor and the best adviser we could have asked for. Their advice, encouragement, and constructive feedback served as a source of innovative ideas and inspiration, and were central to the successful completion of this project. The confidence shown in us was the biggest source of motivation throughout this journey.

We extend our sincere thanks to the Edunet Foundation, Microsoft, LinkedIn, and SAP for providing this fellowship opportunity, the learning resources, and the platform to develop and showcase our AI skills.

We are also grateful to our institution, Graphic Era Deemed To Be University, for providing the necessary infrastructure and support throughout the project duration.

Finally, we thank our family and friends for their continuous encouragement and support.

ABSTRACT OF THE PROJECT

This project is part of the AICW (Artificial Intelligence in the Contemporary World) Fellowship Program, a collaborative initiative by Edunet Foundation in partnership with Microsoft, LinkedIn, and SAP. The fellowship is designed to equip students with practical, hands-on skills in Artificial Intelligence, Data Analytics, and Business Intelligence tools.

This project is developed as part of the AICW (Artificial Intelligence in the Contemporary World) Fellowship Program, aimed at enhancing practical skills in data analytics and business intelligence using modern tools.

The project focuses on analyzing Netflix data to understand content performance, viewer engagement, and revenue trends. With the rapid growth of digital streaming platforms, analyzing large volumes of content data has become essential for effective decision- making. However, extracting meaningful insights from such data remains a challenge without proper tools and techniques.

A Netflix dataset containing over 10,000 records and 16 attributes was used for this analysis. The dataset includes information such as genre, ratings, views, revenue, region, and user engagement metrics. Data cleaning and preprocessing were performed to ensure consistency, accuracy, and readiness for analysis.

The cleaned dataset was analyzed using Microsoft Power BI, where interactive dashboards were created to visualize key performance indicators such as total views, revenue, average ratings, and genre-wise performance. Various visualizations including bar charts, line charts, and maps were used to identify trends and patterns.

Additionally, AI tools such as ChatGPT and Microsoft Copilot were utilized to assist in data interpretation, insight generation, and report writing, improving efficiency and analytical depth.

The project successfully demonstrates how business intelligence tools combined with AI can transform raw data into actionable insights. The findings provide valuable information for optimizing content strategies, improving user engagement, and supporting data-driven decision-making in the Netflix industry.

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CHAPTER 1

Introduction

This chapter introduces the project undertaken as part of the AICW Fellowship Program. It outlines the background, the problem being addressed, the motivation behind this work, and the objectives and scope of the project. The fellowship projects span multiple domains including Data Analytics, Business Intelligence, Healthcare Analytics, Content Generation, and AI-assisted Reporting.

1.1 Problem Statement

The rapid growth of OTT platforms such as Netflix and Amazon Prime has resulted in a massive amount of content and user engagement data. However, analyzing this data to understand viewer preferences, content performance, and revenue trends remains a challenge.

Organizations often lack structured insights into which genres, regions, or content types drive higher engagement and profitability. Without proper data visualization and analysis, decision-makers may fail to optimize content strategy and user retention.

This project aims to analyze Netflix data to identify key factors influencing content popularity, user ratings, and revenue generation using Power BI and AI-assisted tools.

1.2 Motivation

This project was chosen due to the increasing relevance of OTT platforms in today's digital entertainment industry. Understanding how data analytics can improve content strategy and user engagement is highly valuable for both business and analytical roles.

Additionally, this project aligns with modern trends in data analytics and business intelligence, where tools like Power BI and AI assistants such as Copilot are used to derive actionable insights from large datasets.

1.3 Objectives

1. To analyze Netflix data containing 10,000+ records and multiple attributes
2. To clean and preprocess the dataset for accurate analysis
3. To identify trends in content popularity, ratings, and user engagement
4. To build interactive dashboards using Power BI
5. To generate business insights using AI tools like Copilot

1.4 Scope of the Project

- This project focuses on analyzing a Netflix dataset containing content details such as genre, ratings, views, and revenue.
- The tools used are Power BI and Copilot for data analysis and visualization.
- The project is limited to descriptive analytics and does not include predictive modelling or real-time data integration.

CHAPTER 2

Literature Survey

A literature survey reviews existing research, tools, and methodologies relevant to the project domain. It helps identify gaps that the current project aims to address.

2.1 Existing Work and Research

With the rapid growth of Over-The-Top (OTT) platforms such as Netflix, Amazon Prime, and Disney+, there has been a significant increase in the volume of digital content and user engagement data. Researchers and analysts have focused on understanding viewer behavior, content performance, and recommendation systems using data analytics and machine learning techniques.

Several studies highlight the importance of analyzing user preferences, ratings, and viewing patterns to improve content recommendations and enhance user experience. Business Intelligence tools like Power BI and Tableau have been widely used to visualize OTT data and generate actionable insights for decision-makers.

Existing approaches primarily focus on:

- Recommendation systems using machine learning algorithms
- Viewer segmentation based on behavior and demographics
- Content performance analysis using ratings and engagement metrics

However, many of these solutions require advanced technical expertise and complex models, making them less accessible for beginners or non-technical users.

This project focuses on a simplified and practical approach using Power BI and AI tools to analyze Netflix data and generate insights without relying on complex machine learning models.

2.2 Tools and Technologies Review

- Microsoft Excel — Microsoft Excel is used for initial data handling, cleaning, and preprocessing. Features such as sorting, filtering, conditional formatting, and basic formulas help in preparing the dataset before visualization.
- Microsoft Power BI — Microsoft Power BI is a powerful business analytics tool that enables users to create interactive dashboards and visualizations. It allows users to connect multiple data sources, transform data, and generate insights using DAX (Data Analysis Expressions).
  Power BI is widely used in industry for decision-making and reporting.
- Microsoft Copilot (AI) — Microsoft Copilot is an AI-powered assistant integrated with Microsoft tools. It helps automate tasks, generate summaries, and provide insights using natural language prompts.
  In this project, Copilot assists in: Data understanding, Insight generation, Report writing
- AI Chatbots (e.g., ChatGPT, Gemini) — AI chatbots are used for prompt-based analysis, content generation, and explanation of insights. These tools enhance productivity and reduce manual effort in analysis and documentation.

2.3 Gaps Identified

Despite the availability of advanced research and tools in OTT analytics, several gaps still exist:

- Most existing solutions rely on complex machine learning models, which are difficult for beginners to implement
- Limited focus on using simple tools like Power BI for Netflix data analysis
- Lack of integration between AI tools (Copilot/ChatGPT) and business intelligence workflows
- Insufficient emphasis on storytelling and visualization for non-technical stakeholders

This project addresses these gaps by providing a simplified, tool-based approach using Power BI and AI tools to analyze Netflix data effectively.

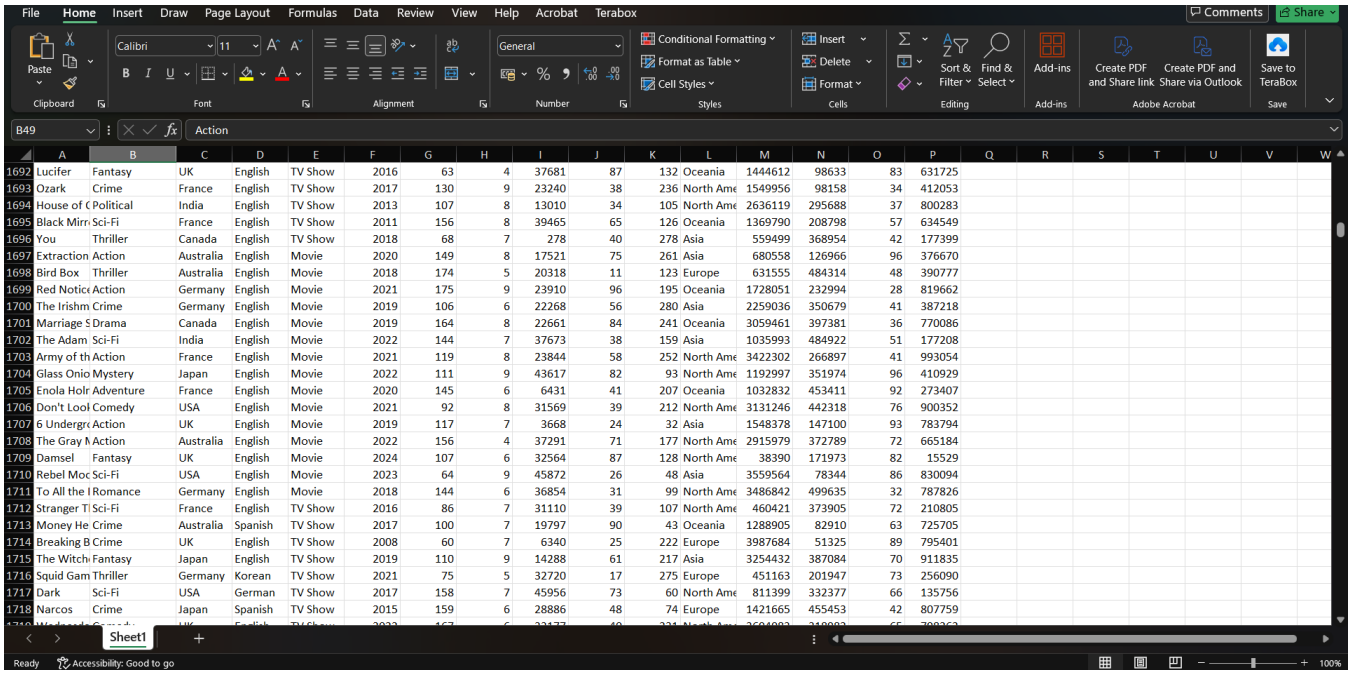
CHAPTER 3

Proposed Methodology

This chapter describes the methodology adopted to achieve the project objectives. It covers the system design, modules used, data flow, advantages, and requirement specifications.

3.1 System Design

- Data Source / Input: CSV file with 10000 rows and 16 columns
- Data Cleaning: Removed inconsistencies, handled missing values, standardized formats
- Analysis / Processing: Used Power BI visuals and DAX measures
- Visualization / Output: Interactive dashboard showing content performance and engagement.



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1694	House of (	Political	India	English	TV Show	2013	107	8	13010	34	105	North Ame	2636119	295688	37	800283
1695	Black Mirr	Sci-Fi	France	English	TV Show	2011	156	8	39465	65	126	Oceania	1369790	208798	57	634549
1696	You	Thriller	Canada	English	TV Show	2018	68	7	278	40	278	Asia	559499	368954	42	177399
1697	Extraction	Action	Australia	English	Movie	2020	149	8	17521	75	261	Asia	680558	126966	96	376670
1698	Bird Box	Thriller	Australia	English	Movie	2018	174	5	20318	11	123	Europe	631555	484314	48	390777
1699	Red Notic	Action	Germany	English	Movie	2021	175	9	23910	96	195	Oceania	1728051	232994	28	819662
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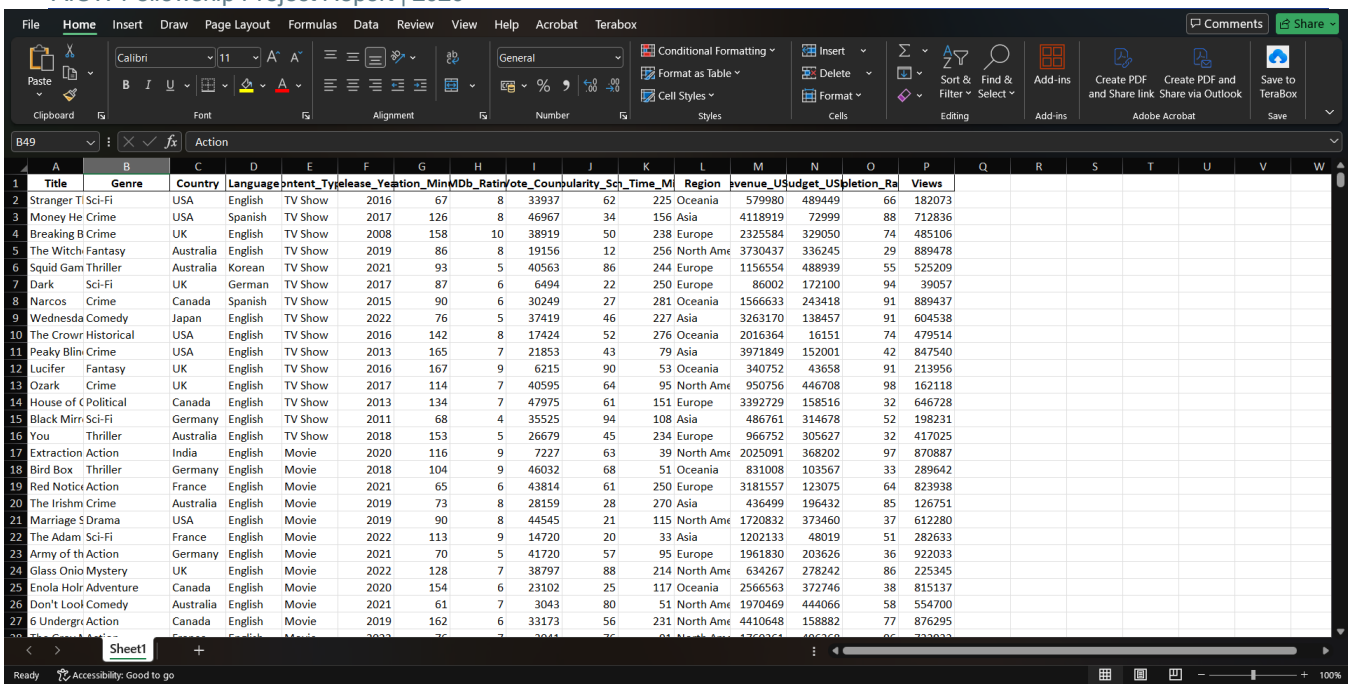
1702	The Adam	Sci-Fi	India	English	Movie	2022	144	7	37673	38	159	Asia	1035993	484922	51	177208
1703	Army of th	Action	France	English	Movie	2021	119	8	23844	58	252	North Ame	3422302	266897	41	993054
1704	Glass Onio	Mystery	Japan	English	Movie	2022	111	9	43617	82	93	North Ame	1192997	351974	96	410929
1705	Enola Holr	Adventure	France	English	Movie	2020	145	6	6431	41	207	Oceania	1032832	453411	92	273407
1706	Don't Lool	Comedy	USA	English	Movie	2021	92	8	31569	39	212	North Ame	3131246	442318	76	900352
1707	6 Undergr	Action	UK	English	Movie	2019	117	7	3668	24	32	Asia	1548378	147100	93	783794
1708	The Gray M	Action	Australia	English	Movie	2022	156	4	37291	71	177	North Ame	2915979	372789	72	665184
1709	Damsel	Fantasy	UK	English	Movie	2024	107	6	32564	87	128	North Ame	38390	171973	82	15529
1710	Rebel Moc	Sci-Fi	USA	English	Movie	2023	64	9	45872	26	48	Asia	3559564	78344	86	830094
1711	To All the I	Romance	Germany	English	Movie	2018	144	6	36854	31	99	North Ame	3486842	499635	32	787826
1712	Stranger T	Sci-Fi	France	English	TV Show	2016	86	7	31110	39	107	North Ame	460421	373905	72	210805
1713	Money He	Crime	Australia	Spanish	TV Show	2017	100	7	19797	90	43	Oceania	1288905	82910	63	725705
1714	Breaking B	Crime	UK	English	TV Show	2008	60	7	6340	25	222	Europe	3987684	51325	89	795401
1715	The Witch	Fantasy	Japan	English	TV Show	2019	110	9	14288	61	217	Asia	3254432	387084	70	911835
1716	Squid Gam	Thriller	Germany	Korean	TV Show	2021	75	5	32720	17	275	Europe	451163	201947	73	256090
1717	Dark	Sci-Fi	USA	German	TV Show	2017	158	7	45956	73	60	North Ame	811399	332377	66	135756
1718	Narcos	Crime	Japan	Spanish	TV Show	2015	159	6	28886	48	74	Europe	1421665	455453	42	807759

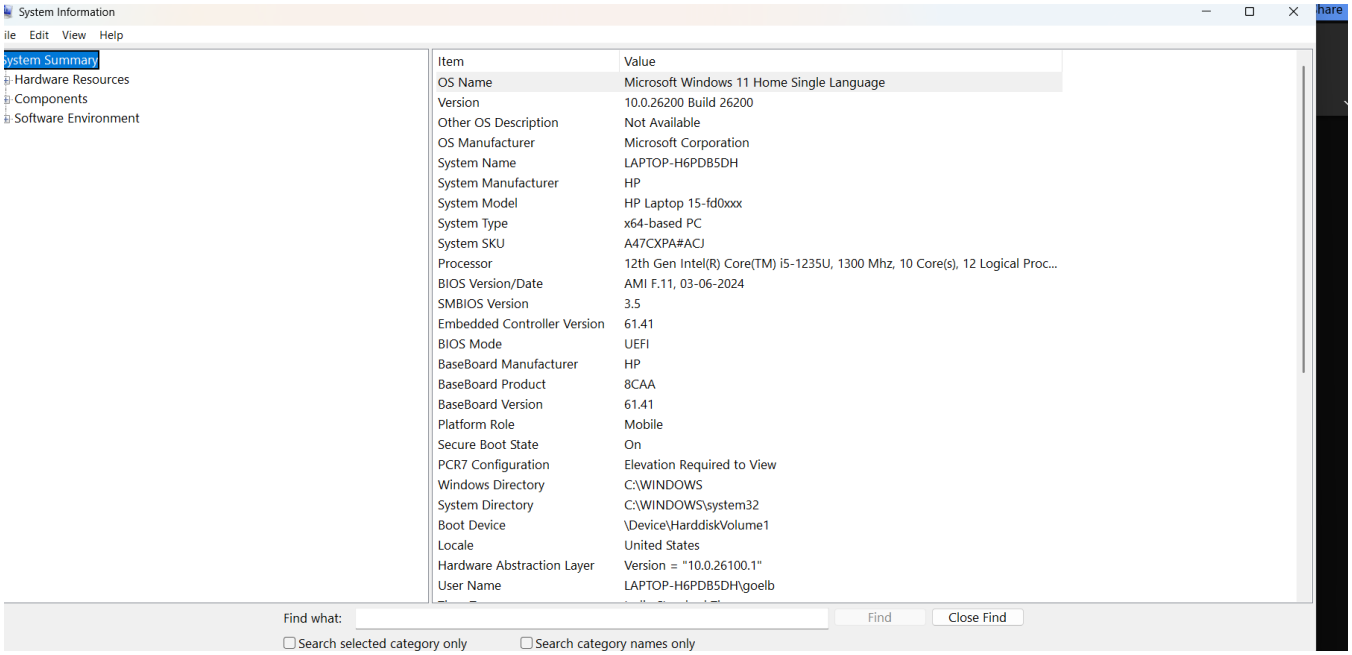


Title	Genre	Country	Language	ontent_Ty	elease_Ye	ation_Min	MDb_Ratin	ote_Coun	ularity_Sc	_Time_M	Region	evenue_US	udget_US	pletion_Ra	Views
Stranger Ti	Sci-Fi	USA	English	TV Show	2016	67	8	33937	62	225	Oceania	579980	489449	66	182073
Money He	Crime	USA	Spanish	TV Show	2017	126	8	46967	34	156	Asia	4118919	72999	88	712836
Breaking B	Crime	UK	English	TV Show	2008	158	10	38919	50	238	Europe	2325584	329050	74	485106
The Witch	Fantasy	Australia	English	TV Show	2019	86	8	19156	12	256	North Ame	3730437	336245	29	889478
Squid Gam	Thriller	Australia	Korean	TV Show	2021	93	5	40563	86	244	Europe	1156554	488939	55	525209
Dark	Sci-Fi	UK	German	TV Show	2017	87	6	6494	22	250	Europe	86002	172100	94	39057
Narcos	Crime	Canada	Spanish	TV Show	2015	90	6	30249	27	281	Oceania	1566633	243418	91	889437
Wednesda	Comedy	Japan	English	TV Show	2022	76	5	37419	46	227	Asia	3263170	138457	91	604538
The Crown	Historical	USA	English	TV Show	2016	142	8	17424	52	276	Oceania	2016364	16151	74	479514
Peaky Blin	Crime	USA	English	TV Show	2013	165	7	21853	43	79	Asia	3971849	152001	42	847540
Lucifer	Fantasy	UK	English	TV Show	2016	167	9	6215	90	53	Oceania	340752	43658	91	213956
Ozark	Crime	UK	English	TV Show	2017	114	7	40595	64	95	North Ame	950756	446708	98	162118
House of (	Political	Canada	English	TV Show	2013	134	7	47975	61	151	Europe	3392729	158516	32	646728
Black Mirr	Sci-Fi	Germany	English	TV Show	2011	68	4	35525	94	108	Asia	486761	314678	52	198231
You	Thriller	Australia	English	TV Show	2018	153	5	26679	45	234	Europe	966752	305627	32	417025
Extraction	Action	India	English	Movie	2020	116	9	7227	63	39	North Ame	2025091	368202	97	870887
Bird Box	Thriller	Germany	English	Movie	2018	104	9	46032	68	51	Oceania	831008	103567	33	289642
Red Notice	Action	France	English	Movie	2021	65	6	43814	61	250	Europe	3181557	123075	64	823938
The Irishm	Crime	Australia	English	Movie	2019	73	8	28159	28	270	Asia	436499	196432	85	126751
Marriage S	Drama	USA	English	Movie	2019	90	8	44545	21	115	North Ame	1720832	373460	37	612280
The Adam	Sci-Fi	France	English	Movie	2022	113	9	14720	20	33	Asia	1202133	48019	51	282633
Army of th	Action	Germany	English	Movie	2021	70	5	41720	57	95	Europe	1961830	203626	36	922033
Glass Onio	Mystery	UK	English	Movie	2022	128	7	38797	88	214	North Ame	634267	278242	86	225345
Enola Holr	Adventure	Canada	English	Movie	2020	154	6	23102	25	117	Oceania	2566563	372746	38	815137
Don't Look	Comedy	Australia	English	Movie	2021	61	7	3043	80	51	North Ame	1970469	444066	58	554700
6 Undergr	Action	Canada	English	Movie	2019	162	6	33173	56	231	North Ame	4410648	158882	77	876295

3.2 Modules / Components Used

List the key modules or functional components of your project:

- Module 1 – Data Ingestion: Loading the raw dataset into Power BI.
- Module 2 – Data Cleaning: Removing inconsistencies, duplicates, and null values.
- Module 3 – Data Analysis: Applying calculations, formulas, AI prompts, or DAX measures.
- Module 4 – Visualization / Report Generation: Building dashboards, charts, or AI-generated reports.
- Module 5 – AI Integration: Using Copilot for analysis, content generation, or insight summaries.



Item	Value
OS Name	Microsoft Windows 11 Home Single Language
Version	10.0.26200 Build 26200
Other OS Description	Not Available
OS Manufacturer	Microsoft Corporation
System Name	LAPTOP-H6PDB5DH
System Manufacturer	HP
System Model	HP Laptop 15-fd0xxx
System Type	x64-based PC
System SKU	A47CXPA#ACJ
Processor	12th Gen Intel(R) Core(TM) i5-1235U, 1300 Mhz, 10 Core(s), 12 Logical Proc...
BIOS Version/Date	AMI F.11, 03-06-2024
SMBIOS Version	3.5
Embedded Controller Version	61.41
BIOS Mode	UEFI
BaseBoard Manufacturer	HP
BaseBoard Product	8CAA
BaseBoard Version	61.41
Platform Role	Mobile
Secure Boot State	On
PCR7 Configuration	Elevation Required to View
Windows Directory	C:\WINDOWS
System Directory	C:\WINDOWS\system32
Boot Device	\Device\HarddiskVolume1
Locale	United States
Hardware Abstraction Layer	Version = "10.0.26100.1"
User Name	LAPTOP-H6PDB5DH\goelb

3.3 Data Flow Diagram (DFD)

A Data Flow Diagram (DFD) is a graphical representation of how data flows through your system. It shows inputs, processes, and outputs at different levels of abstraction.

DFD Level 0 — Context Diagram

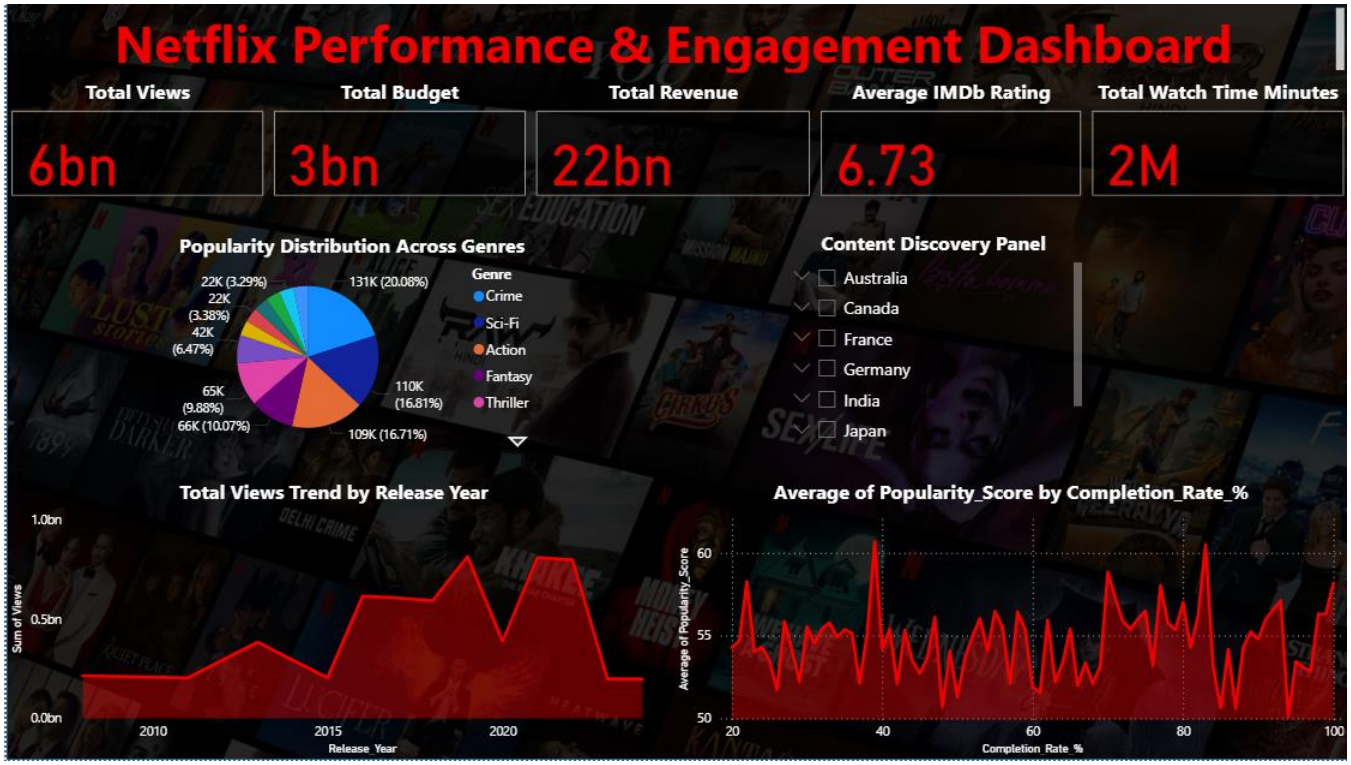
The system takes raw Netflix dataset as input and produces cleaned data and visual dashboard as output.

Raw Dataset → Netflix Analytics System → Dashboard & Insights

DFD Level 1 — Detailed Process Flow

- Sub-process 1: Data Loading and Validation
- Sub-process 2: Data Cleaning and Transformation
- Sub-process 3: Analysis and AI Processing
- Sub-process 4: Visualization and Report Generation

Dataset → Cleaning → Analysis → Visualization → Insights



3.4 Advantages

- Efficient handling of large datasets (10,000+ records)
- Provides clear and interactive visual insights
- Improves decision-making using data-driven approach
- Integrates AI tools for faster and better analysis
- Scalable and adaptable to real-world OTT platforms
- User-friendly and suitable for non-technical users

3.5 Requirement Specification

Hardware Requirements

Component	Specification
Processor	Intel Core i5
RAM	8 GB
Storage	Minimum 50 GB free disk space
Display	1366 x 768 resolution or higher
Internet	Required for Copilot access

Software Requirements

Software	Details
Operating System	Windows 11
Microsoft Excel	Microsoft 365 (Excel with Copilot)
Power BI Desktop	Latest version
Microsoft Copilot	Integrated via Microsoft 365 subscription
AI Chatbot	Microsoft Copilot (web)
Web Browser	Google Chrome

CHAPTER 4

Implementation and Results

This chapter presents the actual implementation of the project and the results obtained at each stage. Screenshots of dashboards, cleaned datasets, or generated content should be inserted where indicated.

4.1 Data Cleaning Results

Describe the data cleaning process and present the before/after comparison of your dataset.

- Dataset source: Dataset is initially taken from Kaggle and later necessary changes are made for analytical purposes
- Original dataset size: 10500 rows and 18 columns
- Issues found: □ Inconsistent formatting in categorical columns (Genre, Country, Language)
  □ Presence of duplicate-like patterns in generated data Variation
  □ in numerical ranges (ratings, revenue, views) Column naming
  □ inconsistencies
- Cleaning steps applied: □ Standardized column names and formats
  □ Ensured consistent categorical values
  □ Verified data types (numerical and categorical separation)
  □ Removed unnecessary inconsistencies
  □ Checked for missing/null values
- Cleaned dataset size: 10,000 rows and 16 columns

4.2 Visualization / Output Results

Present the key visualizations, dashboards, or generated content produced by your project.

- Dashboard / Chart 1: **Genre-wise Content Analysis**

  A bar chart was used to display the distribution of content across different genres. It helped identify the most popular genres on the platform.
- Dashboard / Chart 2: **Year-wise Content Release Trend**

  A line chart was used to show how content production has evolved over time. This highlight growth trends in OTT platforms.
- Dashboard / Chart 3: **Average Popularity Score by completion rate**

  A stacked area chart was used to compare Average popularity score with completion rate. This helps understand audience perception vs critic ratings.

4.3 AI Tool Integration Results

Describe how the AI tool (Copilot or Chatbot) was used in your project and what results it produced.

- Tool used: Microsoft Copilot, ChatGPT
- Prompts used: “Analyze Netflix dataset and identify top-performing genres”
  “Provide insights on relationship between ratings and views”

"Summarize key business insights from Netflix performance dashboard"

- Output quality: The AI tools generated clear, structured, and relevant insights which improved the overall quality of analysis.
- How it enhanced the project: Reduced manual effort in analysis
  Faster insight generation
  Improved report writing and presentation
  Enhanced understanding of data patterns

CHAPTER 5

Discussion and Conclusion

5.1 Key Findings

The analysis of the Netflix dataset revealed several important insights:

- Crime and Sci-fi genres showed the highest viewer engagement in terms of popularity score.
- Content with higher IMDb ratings generally attracted more users and had better completion rates
- Certain countries, especially USA and India, contributed significantly to total revenue
- Popularity score and user ratings were strong indicators of content success
- TV Shows had higher average watch time compared to Movies, indicating stronger user retention

These findings highlight the importance of data-driven strategies in optimizing Netflix platform performance.

5.2 Project Links

Share the relevant links to your project work for evaluation.

- GitHub Link of the Project: [Paste your GitHub repository link here]
- Video Recording of Project Demonstration: https://drive.google.com/file/d/13iDAZJyGVNjd80vQ1_F8YnyyCTDY8RfH/view?usp=drive_link

Netflix Project.pbix

- Power BI / Dashboard Link :

5.3 Limitations

Despite successful implementation, the project has certain limitations:

- The dataset used is taken from Kaggle and later some necessary changes and may not fully represent real-world Netflix data
- The analysis is limited to descriptive insights and does not include predictive modeling
- Real-time data integration was not implemented
- AI tool usage was limited to basic prompts and analysis

5.4 Future Work

The project can be further enhanced in the following ways:

- Integration of real-time Netflix data using APIs
- Implementation of machine learning models for content recommendation and forecasting
- Development of a web-based dashboard for broader accessibility
- Inclusion of user demographic data for deeper analysis

5.5 Conclusion

This project successfully demonstrates how Power BI and AI tools can be used to analyze large-scale Netflix datasets. The dashboard provides meaningful insights into content performance, user engagement, and revenue trends.

All project objectives were achieved, and the use of AI tools significantly improved efficiency and analytical depth. This project highlights the importance of data-driven decision-making in the entertainment industry.

REFERENCES

[1] Microsoft Corporation. Microsoft Excel Documentation. Available at: https://support.microsoft.com/en-us/excel

[2] Microsoft Corporation. Power BI Documentation. Available at: https://learn.microsoft.com/en-us/power-bi/

[3] Microsoft Corporation. Microsoft Copilot – AI-Powered Productivity. Available at: https://copilot.microsoft.com

[4] Edunet Foundation. AICW Fellowship Program Guidelines. 2026.

[5] Kaggle. *OTT / Netflix Dataset Resources.*
Available at: https://www.kaggle.com/

APPENDICES

Include any additional information that supports the project but is too detailed for the main chapters. This may include:

Appendix A — Sample Dataset

Appendix B — Key Excel Formulas / DAX Measures Used

List any important formulas, functions, or DAX measures used in the project:

- =IFERROR(VLOOKUP(...), "Not Found") – used for data validation
- =COUNTIF(range, criteria) – used to count duplicates
- Power BI DAX: Total Sales = SUM(Sales[Revenue]) – used in dashboard KPI card

Appendix C — AI Prompts Used

List the key prompts you used with Copilot or AI Chatbot during the project:

- Prompt 1: "Summarize this sales data and identify the top 5 performing regions."
- Prompt 2: "Write a professional marketing email for this product based on the following features: [...]"
- Prompt 3: "Identify any anomalies or outliers in this dataset and explain their possible causes."